The 2026 Raleigh Division B Invitational, Presented by NCBioNetwork

Crime Busters

Test Booklet: Division B

DO NOT WRITE ON THIS TEST

Open this booklet when the event leader starts time

Section A - Background

The Crime Scene

The murder of Dr. Dihydro Monoxide, CEO of the top water distributor, was supposed to be swept under the rug, but it wasn't long until reporters swarmed the estate with never-ending questions. The autopsy showed he was strangled! By who nobody knows, but it's your job to figure out!

Suspect One - His Wife

While his wife, Mrs. Carbana Monoxide, was extremely impulsive at times and defined by her close relationship with the family cat, she swears up and down she could never kill anyone! She claims she was baking brownies at her sister's at the time of the murder.

Suspect Two - His Rival

His rival, Agua de Fina, certainly has the motive to murder his greatest rival he claims he was at the beach taking a "much needed" vacation at the time of the murder.

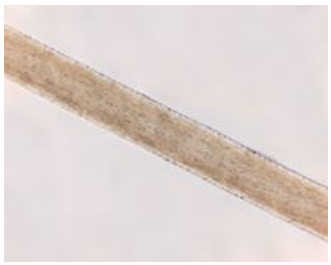
Suspect Three - His Son

His son, Dasan, always knew that when his father passed away, he'd inherit the company and be able to claim a good portion of his father's fortune, with the rest going to his wife. Having just turned 18, he expresses a great deal of concern for his future and feels restless when it comes to his future as a Monoxide. He claims to have been working on his chemistry homework and experimenting with common reactions.

Section B - Identification of Substances

Powders			
Identification	*Name* (2 Points each)	*Formula* (2 Points each)	*Who does this point to* (1 Point each)
S1			
S2			
S3			
S4			
S5			
S6			
S7			
S8			
S9			
Liquids			
Identification	*Name* (2 Points each)	*Formula* (2 Points each)	*Who does this point to* (1 Point each)
L1			
L2			
L3			

Section C - Polymer Testing

Plastics				
Identification	*Description/Test Results* (1 Point each)	*Plastic Name* (2 Points each)	*Formula* (2 Points each)	*Who does this point to?* (1 Point each)
P1				
P2				
Hair				
Identification	*Picture*	*Hair Name* (3 Points each)	*Who does this point to?* (1 Point each)	
H1				
H2				

Fibers			
Identification	*Cloth Microscope Image*	*Fiber Name* (3 Points)	*Who does this point to?* (1 Points)
C1			

Section D - Chromatography

Both Dasan and Mrs. Monoxide have a “signature” pen they like to use. Dasan uses it to take notes on finances, while Mrs. Monoxide focuses more on inks that can survive in the kitchen. A chromatography test was done on both pens.



Pen Found at the scene



Dasan’s Pen

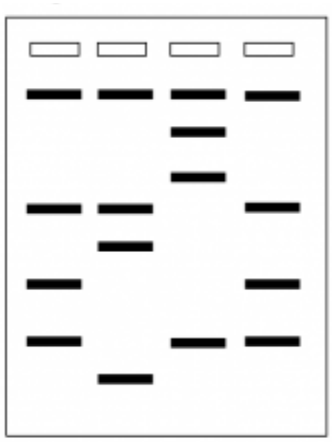


Mrs. Monoxide’s Pen

Who’s pen is it? (5 Points)

Section E - Physical Evidence

DNA was found on the victims neck. Find out whose DNA it was.



1 2 3 4

1) Crime Scene
2) Dasan
3) Agua de Fina
4) Mrs. Monoxide

Who's DNA is this? (5 Points)

Section F - Analysis

1. Who committed the crime? (10 Points)

2. Analyze your evidence pointing to them and why they did it. (30 Points)